



Mehrdad Baratian

Ph.D. Candidate
Tulane University, New Orleans

✉ Mbaratian@tulane.edu

🌐 <https://github.com/M-Baratian>

🌐 www.linkedin.com/in/m-baratian

EDUCATION

- **Tulane University** 2021-present
*Ph.D. Candidate in **Econometrics and Quantitative Economics***
Specializations: Applied Microeconomics, Health Economics, Innovation
New Orleans, LA.
- **Tulane University** 2021-2023
*Master in **Econometrics and Quantitative Economics***
Specializations: Applied Microeconomics
New Orleans, LA.
- **Tehran Polytechnic** 2016-2018
*Master of Science: **Business Administration***
Graduated Top 3 in Class
Tehran
- **University of Isfahan** 2010-2015
*Bachelor of Science: **Mechanical Engineering***
Achieved a Top 1% Ranking in the National University Entrance Exam
Isfahan

FIELDS

Health Economics, Innovation, Labor Economics

WORKING PAPERS

- **Shocks, Reallocation, and the Direction of Innovation: The Case of COVID-19** *Job Market Paper*
 - **Abstract:** Shocks reallocate attention, lab capacity, and funding across fields. I ask how major shocks reshape the field composition of research effort and whether substitution or complementarity dominates. Using COVID-19 as a global shock, I map PubMed articles (2017–2022) into 30 topical clusters with NLP and construct monthly research-effort indices. I then separate level shifts from trend breaks using interrupted time series and a stacked event-study difference-in-differences. Three patterns emerge. First, trend slowdowns: relative to pre-period growth, the median cluster's monthly growth rate falls by about seven percentage points, with many clusters registering statistically significant declines. Second, substitution toward the shock-proximate area: negatively exposed fields experience within-cluster effort drops on the order of 10–13% relative to their counterfactual pre-trends, while the COVID-19 cluster's share of total effort rises sharply at the peak. Third, complementarities: a subset of method- or platform-adjacent clusters accelerates by roughly 4–6 percentage points. These findings imply that policymakers should prioritize neglected clusters during shocks—via bridge funding and access safeguards—to mitigate unintended crowd-out, while also supporting complementary platforms that scale beneficial spillovers. Beyond COVID-19, the measurement and identification strategy provides a general framework to characterize the structure of reallocation under any major shock, guiding targeted interventions when disruptions arise.
- **SNAP Generosity and Mental Health** *Working Paper*
 - Methods: Difference-in-Differences; dose-response Difference-in-Differences; XGBoost machine learning for data imputation
 - Finding: SNAP allotments support mental health during economic hardship, with effects emerging beyond a minimum benefit threshold.

TEACHING EXPERIENCES

- **Research & Teaching Assistant, Tulane University** 2022-present
 - Courses: Econometrics II (Ph.D. Level); Machine Learning and Econometrics; Intermediate Microeconomics; Health Economics; Introduction to Macroeconomics; Introduction to Microeconomics.
- **Teaching Assistant — Strategic Management, Tehran Polytechnic** Spring 2018
 - Course: Strategic Management.

AWARDS

- APPAM Excellence, Equity and Inclusion Student Fellowship 2026–2027
- Cato University Summer Term II (Full Scholarship), Cato Institute August 2026
- J.E. Land Fund for School of Liberal Arts Graduate Student Travel July 2026
- Graduate Student Summer Research Award, Tulane University June 2024, May 2025
- Graduate Studies Student Association Travel Award, Tulane University May 2024
- Joachim-Herz Foundation Award June 2024
- Connolly Alexander Institute for Data Science Research Grant Award June 2024, April 2025

CONFERENCE AND PRESENTATION

• First Author Presenter

- Southern Economic Association (SEA) | Houston, TX. *November 2026*
- Association for Public Policy Analysis and Management (APPAM) | Boston, MA. *November 2026*
- American Society of Health Economists (ASHEcon) | Minneapolis, MN. *June 2026*
- Association for Public Policy Analysis and Management (APPAM) | Seattle, WA. *Fall 2025*
- Southern Economic Association (SEA) | Tampa, FL. *Fall 2025*
- Association for Public Policy Analysis and Management (APPAM) | Washington, D.C. *Fall 2024*
- Essen Economics of Mental Health Workshop, CINCH - Health Economics Research Center | Universität Duisburg-Essen. Minneapolis, MN. *August 2024*
- American Society of Health Economists (ASHEcon) | San Diego, CA. *June 2024*

PEER REVIEW SERVICE

• Conference Referee

- Association for Public Policy Analysis and Management (APPAM) Fall Research Conference *Fall 2025*

• Journal Referee

- Journal of Data Analytics and Intelligent Decision-Making *Fall 2025*

WORKING EXPERIENCE

Statistical Research Consultant, CAIDS, Tulane University: *Fall 2026*

- Consult with Tulane faculty, staff, and student researchers at the Connolly Alexander Institute for Data Science on statistical methodology, research design, and analysis, with support in Python, R, and Stata.

Leadership Role at Tulane University: *2022-2024*

- Vice President, Graduate Student Association—led cross-council initiatives; managed budgets; university liaison.

Negin Bazr Co. *2018-2021*

- Product Marketing Manager, Market Researcher.
- Developed and demonstrated advanced teamwork, decision-making, critical thinking, and problem-solving skills within the private sector, contributing to successful project outcomes and driving organizational efficiency.

TECHNICAL SKILLS

- **Causal Inference:** Difference-in-Differences; Regression Discontinuity; Matching; Instrumental Variables
- **Quantitative Methods:** Econometric Modeling; Predictive Analytics; Deep Learning; Data Analysis
- **Deep Learning / NLP:** Neural Networks; BERT models; TF-IDF; Clustering; Classification
- **Programming:** Python; Stata; SQL
- **Computing:** Amazon Web Services (AWS); Tulane University High-Performance Computing (Cypress)
- **Languages:** English; Farsi

REFERENCES

1- Kevin Callison Associate Professor (Dissertation Chair) Department of Health Policy and Management Tulane University Email: kcallison@tulane.edu Phone: (614) 989-1764	2- Katy Bergstrom Assistant Professor Department of Economics Tulane University Email: kbergstrom@tulane.edu	3- Jonathan Holmes Assistant Professor Department of Economics Tulane University Email: jholmes13@tulane.edu
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